

SAHARSH PATHAK

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B.Tech ECE · 2025–2029



PROFILE

Electronics & Communication Engineering student focused on **AI systems, Edge AI, and open-source LLMs**. Builds and deploys end-to-end intelligent systems spanning local LLM inference, RAG, speech AI, computer vision, embedded hardware, Android, and backend infrastructure.

EDUCATION

B.Tech. Electronics & Communication Engineering
Jaypee Institute of Information Technology (JIIT)

Jul 2025 – Jul 2029 | Noida

PROJECTS & ACHIEVEMENTS

MistMinds — Proactive Edge-AI Air Quality System [GitHub]
ESP32 · PM2.5 · DHT22 · Firebase Realtime Database · React · Vite

2026 | Hackathon

- **1st Place — Climate Data Hackathon 2026**, Rs. 1,00,000 prize; **IoTHON 2026 @ GCET Finalist**.
- Engineered the complete software stack for a **4-member team**, from PM2.5/DHT22 acquisition and edge-side AQI computation to Firebase telemetry and live web visualization.
- Implemented a **closed-loop pollution-control system** that automatically activates a sprinkler when AQI crosses a defined threshold, enabling proactive rather than reactive intervention.
- Deployed the complete working system on **JIIT's 5G infrastructure**.

SEHAT — Smart Edge Healthcare System [Portal] [App]
Raspberry Pi 5 (8GB) · Kotlin · Python · FastAPI · ONNX Runtime · INT8 · IndicConformer · Kokoro TTS

2026 | JIIT

- **Qualified for the Smart India Hackathon internal round**.
- Developed the core software stack for a **6-member team**, spanning Android, backend, AI/speech pipelines, data handling and system integration.
- Engineered the edge pipeline for **temperature, BPM and SpO₂** acquisition, multilingual STT/TTS and offline-first operation with automatic synchronization when connectivity returns.
- Optimized on-device speech inference through model and quantization evaluation; deployed **INT8 inference** on Raspberry Pi 5 (8GB) with Hindi, English and Marathi support.
- Deployed the working system on **JIIT's 5G infrastructure**.

Adaptive Multi-Model RAG Assistant [GitHub]
Python · FastAPI · ChromaDB · Ollama · RAG · Embeddings · Local LLMs

2025–2026 | Open Source

- Engineered a fully local RAG system to evaluate open-source LLMs for practical work-related tasks on consumer hardware, keeping inference and document data within a private local pipeline.
- Implemented adaptive model routing between lightweight and larger local models to balance reasoning capability with local compute requirements.
- Built graph-aware retrieval over an Obsidian knowledge base, expanding context through linked notes alongside vector search and source-grounded generation.

AgriCar / Agribot — Edge-AI Agriculture System [GitHub]
ESP32 · ESP32-CAM · OpenCV · Computer Vision · Embedded Systems · Web

2026 | JIIT

- Integrated the complete robotic system including vehicle movement, control, camera data acquisition, web portal and edge-to-portal communication.
- Implemented and integrated a pretrained OpenCV crop/weed detection pipeline, enabling the system to distinguish weeds from crops and perform **selective pesticide application**.
- Built the camera-to-web data pipeline for real-time visualization and deployed the working system on **JIIT's 5G infrastructure**.

Team Vyomagami — Chassis & Structure Lead
Robotics / Rover Team · CAD · Chassis Engineering

2026

- Prepared CAD designs for the rover chassis, coordinated structural integration with the **bogie and suspension leads**, and planned hardware/material requirements for fabrication.

ADDITIONAL ACHIEVEMENTS & ACTIVITIES

- **Climate Data Hackathon 2026 Winner** (Rs. 1,00,000) · **IoTHON @ GCET Finalist** · **SIH Internal Round Qualifier** (SEHAT) · Member, Open Source Developer Community (OSDC @JIIT).

TECHNICAL SKILLS

LANGUAGES	AI / ML	TOOLS & PLATFORMS	EDGE / HARDWARE
Python · Kotlin · SQL Bash · TypeScript	LLMs · RAG · Embeddings Vector Search · Local Inference Speech AI · Computer Vision Multimodal AI · Quantization	FastAPI · ChromaDB · Ollama · vLLM ONNX Runtime · Git · GitHub · Docker Linux/WSL · Firebase · Vercel · AWS EC2 Android Studio · React · Vite · Unsloth	Raspberry Pi 5 · ESP32 ESP32-CAM · PM2.5 · DHT22 Embedded Systems Robotics · Drones
LinkedIn · GitHub · SEHAT App · SEHAT Portal · MistMinds · AgriCar · Adaptive Multi-Model RAG			